

一、填空题:

1. 记 $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 2 & 1 \end{pmatrix}$, 则 $|A| = D = -2$, 而 $\begin{vmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{vmatrix} = |(A^*)^T| = |A^*| = |A|^2 = 4$

2. 消去变量 z , 得投影柱面 $x^2 + 2y^2 = 16$, 所求投影曲线为 $\begin{cases} x^2 + 2y^2 = 16 \\ z = 0 \end{cases}$

3. $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 3 & 6 \end{array} \right) \xrightarrow{\text{初等行变换}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right)$, 所求坐标是 $(-1, 1, 2)$

4. $n - r(A) = 1$, 又由题设知, $A \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix} = O$, 故所求基础解系是 $\xi = (1, 1, -1, -1)^T$

5. $\because B^{-1} - I$ 的特征值为: $\frac{1}{3} - 1, \frac{1}{4} - 1, \frac{1}{5} - 1, \therefore |B^{-1} - I| = -\frac{2}{5}$

二、选择题:

1. $\because B = (\alpha_1, \alpha_2, \alpha_3) = \begin{pmatrix} 0 & 1 & 0 \\ 5 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$, $\therefore |B| = |A| = \begin{vmatrix} 0 & 1 & 0 \\ 5 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 5$, 选 (B)

2. 选 (D) 3. 选 (A) 4. 选 (A) 5. 选 (A)

三、解: $\because (A + 2E)X = B$, 又 $|A + 2E| = 2 \neq 0$, 故 $A + 2E$ 可逆, $X = (A + 2E)^{-1}B$,

$$\begin{aligned} (A + 2E | B) &= \left(\begin{array}{ccc|cc} 1 & 2 & 3 & 2 & 3 \\ 2 & 2 & 1 & 5 & 2 \\ 3 & 4 & 3 & 4 & 3 \end{array} \right) \rightarrow \left(\begin{array}{ccc|cc} 1 & 2 & 3 & 2 & 3 \\ 0 & -2 & -5 & 1 & -4 \\ 0 & -2 & -6 & -2 & -6 \end{array} \right) \rightarrow \\ &\left(\begin{array}{ccc|cc} 1 & 2 & 3 & 2 & 3 \\ 0 & -2 & -5 & 1 & -4 \\ 0 & 0 & -1 & -3 & -2 \end{array} \right) \rightarrow \left(\begin{array}{ccc|cc} 1 & 2 & 0 & -7 & -3 \\ 0 & -2 & 0 & 16 & 6 \\ 0 & 0 & 1 & 3 & 2 \end{array} \right) \rightarrow \left(\begin{array}{ccc|cc} 1 & 0 & 0 & 9 & 3 \\ 0 & 1 & 0 & -8 & -3 \\ 0 & 0 & 1 & 3 & 2 \end{array} \right) \\ \text{所以, } X &= \begin{pmatrix} 9 & 3 \\ -8 & -3 \\ 3 & 2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{四、解: } (\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5) &= \left(\begin{array}{ccccc} 1 & 1 & 2 & 2 & 1 \\ 0 & 2 & 1 & 5 & -1 \\ 2 & 0 & 3 & -1 & 3 \\ 1 & 1 & 0 & 4 & -1 \end{array} \right) \rightarrow \left(\begin{array}{ccccc} 1 & 1 & 2 & 2 & 1 \\ 0 & 2 & 1 & 5 & -1 \\ 0 & -2 & -1 & -5 & 1 \\ 0 & 0 & -2 & 2 & -2 \end{array} \right) \\ &\rightarrow \left(\begin{array}{ccccc} 1 & 1 & 2 & 2 & 1 \\ 0 & 2 & 1 & 5 & -1 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccccc} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 3 & -1 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \quad (\text{最简阶梯形}) \end{aligned}$$

所以, $r(\alpha_1, \dots, \alpha_5) = 3$, 向量组 $\alpha_1, \dots, \alpha_5$ 线性相关, $\alpha_1, \alpha_2, \alpha_3$ 为其一个极大线性无关组, 且有 $\alpha_4 = \alpha_1 + 3\alpha_2 - \alpha_3$; $\alpha_5 = 0\alpha_1 - \alpha_2 + \alpha_3$.

五、解: $\because \begin{vmatrix} 1 & 1 & 1+\lambda \\ 1 & 1+\lambda & 1 \\ 1+\lambda & 1 & 1 \end{vmatrix} = -\lambda^2(\lambda+3),$

$\therefore$ (1) 当 $\lambda \neq 0$ 且 $\lambda \neq -3$ 时, 方程组有唯一解;

(2) 当 $\lambda = -3$ 时, $r(\tilde{A}) \neq r(A)$, 方程组无解;

(3) 当 $\lambda = 0$ 时, $r(\tilde{A}) = r(A) = 1 < 3$, 方程组有无穷多解, 此时同解方程组为

$$x_1 + x_2 + x_3 = 3, \text{ 故所求通解 } X = k_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}, \text{ 其中 } k_1, k_2 \text{ 为任意实数.}$$

六、解: 过直线 L 的平面束方程: $x + 2y - z + 1 + \lambda(x - y + z - 1) = 0$, 整理得:

$$(1+\lambda)x + (2-\lambda)y + (-1+\lambda)z + 1 - \lambda = 0$$

由题目假设, $(1+\lambda) \cdot 1 + (2-\lambda) \cdot (-1) + (-1+\lambda) \cdot 1 = 0 \Rightarrow \lambda = \frac{2}{3}$

所求平面方程是: $5x + 4y - z + 1 = 0$.

七、解: 二次型对应矩阵为 $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & a \end{pmatrix} \Rightarrow \begin{cases} 5+b=4+a \\ 2a-2=4b \end{cases} \Rightarrow \begin{cases} a=1 \\ b=0 \end{cases}$

$\therefore \lambda_1 = 1, \lambda_2 = 4, \lambda_3 = 0$ 是 A 的三个特征值,

$$A - E = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \text{ 同解方程组为 } \begin{cases} x_1 = -x_2 \\ x_3 = -x_2 \end{cases} \Rightarrow \xi_1 = (-1, 1, -1)^T$$

$$A - 4E = \begin{pmatrix} -3 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \xi_2 = (1, 2, 1)^T$$

$$A - 0E = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \xi_3 = (-1, 0, 1)^T$$

ξ_1, ξ_2, ξ_3 属于不同特征值, 故已经正交, 单位化后, 令 $P = \begin{pmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0 \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{pmatrix}$ 为所求.

八、证明: $(\Rightarrow) \because A$ 是正交矩阵, $\Rightarrow A^T A = E$,

$$\therefore (A\alpha, A\beta) = \alpha^T A^T A \beta = \alpha^T \beta = (\alpha, \beta)$$

$(\Leftarrow)$ 若 $(A\alpha, A\beta) = \alpha^T A^T A \beta = (\alpha, \beta)$, 则 $A^T A = E$, 即 A 为正交矩阵.

一、填空题:

$$1. D_1 = \begin{vmatrix} 3a_{11} & 4a_{11} & -a_{13} \\ 3a_{21} & 4a_{21} & -a_{23} \\ 3a_{31} & 4a_{31} & -a_{33} \end{vmatrix} + \begin{vmatrix} 3a_{11} & -a_{12} & -a_{13} \\ 3a_{21} & -a_{22} & -a_{23} \\ 3a_{31} & -a_{32} & -a_{33} \end{vmatrix} = 0 + 3D = 6$$

$$2. \because A = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot (1, 2, 3), \therefore A^n = 7^{n-1} A$$

3. P 、 Q 均为初等方阵, PA 相当于对 A 做初等行变换, AQ 相当于对 A 做初等列变换,

$$\text{则 } P^{2008} A Q^{2009} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 2007 & 2008 & 2009 \end{pmatrix} Q^{2009} = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 1 & 1 \\ 2009 & 2008 & 2007 \end{pmatrix}$$

$$4. k_1 + k_2 + \cdots + k_r = 1, \quad 5. \text{秩为 } 3 \quad 6. \text{坐标是 } (1, 1, 1) \quad 7. \sqrt{56}$$

$$8. \begin{cases} x^2 + y^2 = 1 \\ z = 0 \end{cases}$$

9. A 有特征值 $\lambda_1 = 2, \lambda_2 = \lambda_3 = 1$, 对于二重特征值 $\lambda = 1$,

$$A - E = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & a & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & 0 \end{pmatrix}, \text{欲使 } 3 - r(A - E) = 2 \Rightarrow a = 0$$

10. $k > 1$ 或 $k < -2$

二、计算下列各题:

1. 解: $\because (A - B)X(A - B) = I$, 又 $|A - B| = 1 \neq 0$, 故 $A - B$ 可逆, $X = [(A - B)^{-1}]^2$,

$$(A - B | I) = \left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$\text{所以, } X = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 2 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$2. \text{解: } \because \begin{vmatrix} 1+\lambda & 1 & 1 \\ 1 & 1+\lambda & 1 \\ 1 & 1 & 1+\lambda \end{vmatrix} = \lambda^2(\lambda+3),$$

$\therefore$ (1) 当 $\lambda = 0$ 时, $r(\tilde{A}) = r(A) = 1 < 3$, 方程组有无穷多解, 此时 β 可由 $\alpha_1, \alpha_2, \alpha_3$ 表示, 且表示法不唯一;

(2) 当 $\lambda = -3$ 时, $r(\tilde{A}) \neq r(A)$, 方程组无解; 此时 β 不能由 $\alpha_1, \alpha_2, \alpha_3$ 表示.

3. 解: 由于 $r(A) = 3$, 故对应齐次线性方程组基础解系只有一个解,

记 $\xi = 2\eta_3 - (\eta_1 + \eta_2) = (5, 4, -3, 6)^T$, 则 ξ 即为对应齐次线性方程组的基础解系, 故所求通解是 $X = k\xi + \eta_3$, 其中 k 为任意实数.

4. 解: 二次型对应矩阵为 $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, $|A - \lambda E| = (-\lambda - 1)^2(-\lambda + 2)$

$\lambda_1 = 2, \lambda_2 = \lambda_3 = -1$ 是 A 的三个特征值,

$$A - 2E = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \xi_1 = (1, 1, 1)^T$$

$$A + E = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \xi_2 = (-1, 1, 0)^T, \quad \xi_3 = (-1, 0, 1)^T,$$

将 ξ_2, ξ_3 正交化, 单位化后, 令 $X = \begin{pmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \end{pmatrix} Y$,

标准形为 $f = 2y_1^2 - y_2^2 - y_3^2$.

三、解: 设所求直线方程为 $L: \frac{x-1}{l} = \frac{y-1}{m} = \frac{z-1}{n}$, 则

$$L \text{ 与 } L_1 \text{ 相交} \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ l & m & n \end{vmatrix} = 0 \Rightarrow l - 2m + n = 0,$$

$$L \text{ 与 } L_2 \text{ 相交} \Rightarrow \begin{vmatrix} 0 & 1 & 1 \\ 2 & 1 & 1 \\ l & m & n \end{vmatrix} = 0 \Rightarrow m - n = 0,$$

所以, $l:m:n = 1:1:1$, $L: \frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}$

四、证明: 设 $k_1\alpha_1 + k_2\alpha_2 = 0$ (1)

两边左乘 A : $A(k_1\alpha_1 + k_2\alpha_2) = k_1\lambda_1\alpha_1 + k_2\lambda_2\alpha_2 = 0$ (2)

(2) $-\lambda_1(1)$: $k_2(\lambda_2 - \lambda_1)\alpha_2 = 0$, $\because \lambda_1 \neq \lambda_2, \alpha_2 \neq 0 \Rightarrow k_2 = 0$,

代入 (1), $\because \alpha_1 \neq 0 \Rightarrow k_1 = 0$, $\therefore \alpha_1, \alpha_2$ 线性无关.

一、填空题:

1. $|A+B| = 2^3(|A|+|B|) = 24$, 2. $\lambda = \cancel{n}$, $\xi = k(1, 1, \dots, 1)^T$ $k \neq 0, k \in R$

3. $\eta = (1, -1, 2, -1)^T$, 4. $X = (1, 1, -1)^T$, 5. $\begin{cases} x^2 + y^2 = 1 \\ z = 0 \end{cases}$

二、选择题:

1. 选 (D) 2. 选 $(\frac{\pi}{3})$ 3. 选 (B) 4. 选 (A) 5. 选 (C) 注: 选项 (C)

中 $\lambda_i > 0$.

三、计算下列各题:

1. 解: $\because (A^{-1} - E)BA = 6A, \Rightarrow B = 6(A^{-1} - E)^{-1}AA^{-1} = 6(A^{-1} - E)^{-1}$,

$$A^{-1} - E = \begin{pmatrix} -\frac{3}{4} & & \\ & 1 & \\ & & -\frac{1}{2} \end{pmatrix} \Rightarrow B = \begin{pmatrix} -8 & & \\ & 6 & \\ & & -12 \end{pmatrix}$$

2. 解: 类似于试卷一的第四题;

3. 解: 类似于试卷一的第五题,

4. 解: (1) $A = \begin{pmatrix} 1-a & 1+a & 0 \\ 1+a & 1-a & 0 \\ 0 & 0 & 2 \end{pmatrix}$, $\because r(A) = 2 \Rightarrow |A| = 0 \Rightarrow a = 0$.

(2) $|A - \lambda E| = \begin{vmatrix} 1-\lambda & 1 & 0 \\ 1 & 1-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{vmatrix} = -\lambda(\lambda-2)^2 \Rightarrow \lambda_1 = 0, \lambda_2 = \lambda_3 = 2$,

当 $\lambda = 0$ 时, $A - 0E = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \xi_1 = (-1, 1, 0)^T$

当 $\lambda = 2$ 时, $A - 2E = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$\Rightarrow \xi_2 = (1, 1, 0)^T, \xi_3 = (0, 0, 1)^T$ (这里 ξ_2, ξ_3 已经正交)

取 $Q = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}$, 经过正交变换 $X = QY$, 标准形为 $f = 2y_2^2 + 2y_3^2$.

(3) $f = 8$ 表示圆柱面.

四、解: 类似于试卷一的第六题

五、解: $\because A(\alpha_1, \alpha_2, \alpha_3) = (A\alpha_1, A\alpha_2, A\alpha_3) = (\alpha_1, 2\alpha_2, 3\alpha_3)$

$$\Rightarrow A = (\alpha_1, 2\alpha_2, 3\alpha_3)(\alpha_1, \alpha_2, \alpha_3)^{-1} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix}$$

一、填空题:

1. 0 2. 线性无关 3. 2 4. $\frac{\pi}{4}$ 5. $\begin{cases} x^2 + y^2 = 1 \\ z = 0 \end{cases}$

二、选择题:

1. 选 (C) 2. 选 (B) 3. 选 (D) 4. 选 (A) 5. 选 (B)

三、解: 按第一列(行)展开, 可得 $D = 7^4 - 9^4$

四、解: $\because X = A^{-1}BA$,

$$(A|E) = \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 \end{array} \right) \rightarrow$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{2} \end{array} \right)$$

$$\text{所以, } X = \begin{pmatrix} 1 & 0 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

五、解: 类似试卷一第四题

六、解: 类似试卷一第五题,

$$\text{七、解: } \bar{v} = \begin{vmatrix} i & j & k \\ 1 & 0 & -4 \\ 2 & -1 & -5 \end{vmatrix} = -4i - 3j - k, \text{ 所求直线为 } \frac{x+3}{4} = \frac{y-2}{3} = \frac{z-5}{1}.$$

$$\text{八、解: 二次型对应矩阵为 } A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & 1 \end{pmatrix} \Rightarrow 2+a=5 \Rightarrow a=3$$

$\therefore \lambda_1 = 0, \lambda_2 = 1, \lambda_3 = 4$ 是 A 的三个特征值,

$$A - 0E = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \xi_3 = (-1, 0, 1)^T$$

$$A - E = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \text{ 同解方程组为 } \begin{cases} x_1 = -x_2 \\ x_3 = -x_2 \end{cases} \Rightarrow \xi_1 = (-1, 1, -1)^T$$

$$A - 4E = \begin{pmatrix} -3 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \xi_2 = (1, 2, 1)^T$$

$$\xi_1, \xi_2, \xi_3 \text{ 属于不同特征值, 故已经正交, 单位化后, 令 } Q = \begin{pmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0 \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{pmatrix} \text{ 为所求,}$$

$f = 4$ 表示椭圆柱面.

九、解： $\because B$ 非零， $\Rightarrow$ 方程组有非零解 $\Rightarrow \begin{vmatrix} 1 & 2 & -2 \\ 2 & -1 & t \\ 3 & 1 & -1 \end{vmatrix} = 0 \Rightarrow t = 1$

又 $A = \begin{pmatrix} 1 & 2 & -2 \\ 2 & -1 & 1 \\ 3 & 1 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow r(A) = 2 \Rightarrow n - r(A) = 1$

故矩阵 B 至多有 1 列线性无关.

